

Jonathan Allen

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EDUCATION

University of Massachusetts Boston

Bachelor of Science in Computer Science

Boston, MA

Sept 2025 — May 2027

- GPA: 3.97, Dean's List

EXPERIENCE

Undergraduate Research Fellow

Jun 2026 — Present

Machine Psychology Lab

Boston, MA

- Built a JavaScript/WebGPU experimentation platform with device profiling, trial orchestration, resumable campaigns, automated validation, and reproducible reporting for browser LLM benchmarking
- Implemented GPU dispatch batching in WebLLM's WebGPU runtime, reducing generation-heavy wall time 35–41% for SmolLM2-360M and 21.1% for Llama 3.2 1B across 660 audited completions with zero failures or output/dispatch mismatches
- Built a crowdsourced browser benchmark with a Cloudflare Workers backend and Airtable storage, collecting participant hardware metadata and inference latency with payload validation and rate limiting
- First author on a manuscript in preparation examining WebGPU runtime optimization for browser-based LLM inference

AI/ML Engineer — IMPACT Practicum

Jan 2026 — May 2026

Venture Development Center

Boston, MA

- Fine-tuned MedSAM 3 and SAM 3 for biotech startup INIA Biosciences, improving held-out mean Dice from 0.62 to 0.97 for segmentation
- Built INIA's Python inference/fine-tuning package with a common API for SAM 3 and MedSAM 3 backends
- Fine-tuned Cellpose-SAM and built a Napari annotation tool with model-assisted pre-labeling for biotech startup Entocellular

Founder & Software Developer

Jan 2026 — Present

Bestella Web Design

Remote

- Built and maintained production web applications for 10+ small-business clients using JavaScript, Eleventy, and Netlify
- Built reusable components and CI/CD workflows; maintained 90+ PageSpeed scores

PROJECTS

Iterated Prisoner's Dilemma Simulation Platform (In Progress), Python, React, TypeScript, FastAPI, PostgreSQL, Docker

- Built an object-oriented Python engine for repeated Prisoner's Dilemma simulations between configurable strategies
- Developing a React/TypeScript interface and FastAPI service for configuring tournaments and visualizing results
- Implementing deterministic simulations, PostgreSQL persistence, aggregate strategy analysis, automated tests, and Docker Compose packaging

C-Grad — Autograd Engine from Scratch in C

- Built a scalar-valued autograd engine in C with forward/backward passes over a dynamic compute graph and backpropagation using topological sorting (add, mul, ReLU)
- Implemented an arena allocator for graph nodes and an MLP training module

SKILLS

- **Languages:** Python, C, C++, Java, JavaScript, TypeScript, SQL, Bash, HTML/CSS
- **ML & Systems:** PyTorch, Scikit-learn, Hugging Face Transformers, Pandas, NumPy, WebGPU, WebLLM
- **Web & Tools:** React, FastAPI, PostgreSQL, Cloudflare Workers, Airtable, Docker, Docker Compose, GitHub Actions, REST APIs, Eleventy, Linux, Git, Netlify